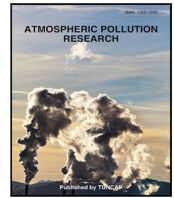




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Modeling PM_{2.5} dispersion from diesel generators in Beirut: Identifying urban hotspots under data scarcity

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ABSTRACT

Diesel generators are emerging as community-initiated solutions to compensate for electricity shortages in cities facing economic crises and infrastructure collapse. In dense urban environments, emissions from these generators pose a significant air quality concern. In the absence of periodic observations from properly distributed sensors, as is the case in Beirut, physically based computational modeling serves as an effective tool for predicting pollutant distribution and assessing mitigation strategies. In this study, we employ the GRAMM-GRAL Lagrangian transport modeling system to simulate PM_{2.5} dispersion originating from diesel generators in Beirut under different meteorological and urban conditions. The simulations account for topography, atmospheric stability, wind speed and direction, building geometry, generator distribution, and exhaust stack heights, referring to the elevation at which generator emissions are released. We evaluate various scenarios by examining overall dispersion patterns, and mean street-level PM_{2.5} concentrations. Using the World Health Organization's AirQ+ tool, we also estimate the potential health impacts based on model-predicted PM_{2.5} concentrations across the city. The results highlight the sensitivity of pollutant dispersion to wind patterns and generator exhaust elevation. Increasing the generator exhaust stack height notably reduces near-ground PM_{2.5} levels, especially under high wind speeds. This study provides a scalable framework for air quality assessment and risk estimation in data-scarce urban environments and informs future urban planning and environmental policy efforts.

1. Introduction

Urban areas with high population density are linked with hotspots of air pollution (Du et al., 2019; Xian et al., 2019). In fact, more than 80% of people living in urban areas are exposed to air quality levels that exceed the World Health Organization (WHO) limits (World Health Organizations, 2016). Particulate matter (PM) refers to a mixture of solid and liquid particles suspended in the air. Fine PM with an aerodynamic diameter of 2.5 µm or less is referred to as PM_{2.5}, and is one of the most critical air pollutants affecting human health (Brunekreef and Holgate, 2002; Kappos et al., 2004; Schwartz et al., 1996); therefore, continuous monitoring of PM_{2.5} concentrations has attracted extensive attention for epidemiological studies in urban areas (Atkinson et al., 2014; Dockery et al., 1993; Fann et al., 2012; Wilson and Suh, 1997). However, many developing cities cannot afford to install and

maintain an expensive air quality monitoring network. Additionally, remote sensing, while valuable, faces inherent limitations in providing detailed PM_{2.5} concentration data (Paciorek and Liu, 2009). Remote sensing methods often suffer from spatial coverage constraints, limiting observations to specific areas and times. While satellite-derived PM_{2.5} products are useful for identifying regional air quality trends, they lack the resolution necessary for capturing street-level variability and informing urban-scale policy decisions (Kloog et al., 2011). These factors collectively contribute to data scarcity in available local-scale observations of PM_{2.5} concentrations.

Consequently, physically based computational modeling stands out as a vital tool for predicting PM_{2.5} concentrations in complex urban environments and investigating various scenarios to develop effective mitigation strategies (Abdallah et al., 2018). Such models, including

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CALPUFF (Scire et al., 2000) and GRAMM-GRAL (Landesregierung, 2018), can simulate pollutant dispersion with high spatial and temporal resolutions, overcoming the limitations of remote sensing and ground-based monitoring networks (Berchet et al., 2017b; Roddis et al., 2017a). Urban morphology further complicates the problem. The abundance of street canyons intensifies air pollution by trapping pollutants and consequently deteriorating air quality, especially when the buildings are tall and the streets are narrow (Liu et al., 2004). Effective pollution hot-spots modeling in such environments, requires high-resolution dispersion models that resolve the buildings that form the street canyons (at a scale $\sim 1-5$ m) (Fabbi et al., 2019; Zhang and Batterman, 2013). The model must also take into account the prevailing wind conditions, the spatial distribution of the emission sources, their emission rates, and the topography.

Among the commonly used modeling approaches, Gaussian plume models and Lagrangian particle dispersion models offer contrasting capabilities in simulating urban air pollution. The Gaussian plume model, while effective in open terrains (Heist et al., 2013), is less suitable for environments with complex street configurations (Di Sabatino et al., 2008). It performs well in accounting for atmospheric stability but cannot simulate interactions with urban structures. In contrast, Lagrangian particle dispersion models rely on three-dimensional, building-resolving flow simulations, enabling a more physically accurate representation of pollutant transport in dense urban domains (Pullen et al., 2005). One such model is the GRAMM (Graz mesoscale model) - GRAL (Graz Lagrangian model) modeling system (Landesregierung, 2018), which couples diagnostic mesoscale wind with Lagrangian particle dispersion modeling. It resolves building-level airflow and dispersion in dense urban topographies. Unlike Gaussian models, it can also capture recirculation zones, flow separation, and street canyon effects with high spatial resolution, making it well suited for identifying urban-scale pollution hot-spots.

Beirut, the capital of Lebanon, lying on the eastern shore of the Mediterranean Sea, experiences high $PM_{2.5}$ concentration levels (Massoud et al., 2011; Saliba et al., 2010) exceeding the WHO limits by 150% (Farah et al., 2018). These high concentrations are linked to long-range pollution transport, such as the dust events originating in the Sahara and Arabian deserts (Lovett et al., 2018), as well as local-range pollution transport due to emissions from traffic (Daher et al., 2013; Massoud et al., 2011), construction sites (Saliba et al., 2010), and diesel generators (Daher et al., 2013). Such long-range transport occurs, because $PM_{2.5}$ particles are small enough to remain airborne and travel long distances under favorable meteorological conditions, unlike their coarser counterparts which settle out quickly.

Among the local sources contributing to these persistent fine particles, diesel generators have drawn particular concern due to their high density across the city, their operation in close proximity to residential areas, and their continuous use during prolonged electricity outages, making them a major contributor to urban air pollution. Diesel generators are widely spread in Beirut and other urban environments as back up energy sources to compensate for electricity outages. These outages result from insufficient generation that does not meet population demand, weak infrastructure for proper transmission and distribution, or rapid industrialization. It is worth noting that several countries in Africa and Asia suffer from both problems (Alam and Powell, 2017). For instance, countries like Syria and Iraq are now relying more on diesel generators to support their grid as a consequence of the recent or ongoing wars (Jassim et al., 2016; NewsDeeply, 2014). In Beirut city, diesel generators are among the major sources of anthropogenic heat, and their contribution to the heat island effect was shown to dominate other factors (Ghadban et al., 2020), and to increase the population's exposure to carcinogens (Daher et al., 2013). A survey conducted in 2017 (Baayoun et al., 2019) on the diesel generators installed in the Hamra neighborhood of Beirut showed that 53% of the buildings were equipped with at least one diesel generator. Additionally, the survey showed that around 469 generators exist in the Hamra area (0.55 km^2).

The daily fuel consumption of diesel generators in the whole city was found to be 747 metric tons on average. This estimate was based on the survey and the imported fuel data provided by the Lebanese Ministry of Energy and Water (MOEW) and the Lebanese Ministry of Environment (MOE).

To the best of our knowledge, very few studies have documented PM emissions or concentration levels from diesel generators in urban environments. One study (Centre for Science and Environment, 2018) conducted in Gurugram, India, showed that diesel generators contribute to about 30%–40% increase in $PM_{2.5}$ levels after three to four hours of operation. Another study done in Nigeria (Oguntoke and Adeyemi, 2017) quantified concentrations of different pollutants including PM at various distances from diesel generators. These studies, however, resort to measurements that are spatially limited, and there are no reported studies in which the dispersion of $PM_{2.5}$ emitted from diesel generators is modeled at the pedestrian level (few meters above ground) over a whole city.

In this work, we address the data scarcity issue by using model-based predictions of $PM_{2.5}$ concentration levels originating from the diesel generators installed in Beirut. These generators are typically equipped with vertical exhaust outlets (referred to as exhaust stacks), whose height varies depending on the installation. Since they are often situated near buildings or within street canyons, the elevation of these exhaust points can significantly influence local pollutant dispersion patterns. The objectives of this study are to (i) gain physical understanding of the impact of and the interplay between the key predictor variables (wind speed and direction, atmospheric stability, and generator exhaust stack height) on the city-scale dispersion patterns, (ii) enhance the epidemiological relevance of our findings by using AirQ+ to perform a comprehensive health risk assessment based on model-predicted $PM_{2.5}$ concentrations under different scenarios, and (iii) assess the efficacy of increasing the generator exhaust stack heights exhausts as a potential mitigation measure. By integrating advanced modeling techniques with considerations of policy implications and long-term sustainability goals, we seek to provide actionable insights for urban policymakers and planners, promote sustainable development, and align with global environmental and health targets.

2. Methodology

2.1. Dispersion model

The GRAMM-GRAL (Version 18.01) modeling system consists of two coupled components that operate at different spatial scales. GRAMM is a non-hydrostatic mesoscale meteorological model that computes wind fields and atmospheric parameters over a broad region encompassing the city and its surroundings (Landesregierung, 2017). GRAL is a microscale Lagrangian particle dispersion model that uses the meteorological fields generated by GRAMM to simulate pollutant transport and dispersion within the urban domain, accounting for the effects of buildings, topography, and emission sources (Landesregierung, 2018). This two-step modeling approach allows detailed simulation of pollutant concentrations at high spatial resolution and is comparable to other modeling suites such as CALPUFF, which comprises CALMET (meteorology), CALPUFF (dispersion), and CALPOST (post-processing) (Scire et al., 2000).

GRAMM-GRAL has been applied to simulate the levels of various pollutants originating from different sources over entire cities in Europe. It was used for evaluation at both the urban and mesoscale levels in Zurich, Laussane (Switzerland), Vienna and Graz (Austria) and Po Valley (Italy) (Almbauer et al., 2000a,b; Berchet et al., 2017a,b; Fabbi et al., 2019; Kurz et al., 2014). Other studies used GRAMM-GRAL to simulate pollution at specific road tunnel portals, highways on complex terrains, specific streets intersections and climate change mitigation in urban mobility (Oettl et al., 2003, 2002; Sturm et al., 2002; Wolking et al., 2018). The performance of this modeling system was evaluated

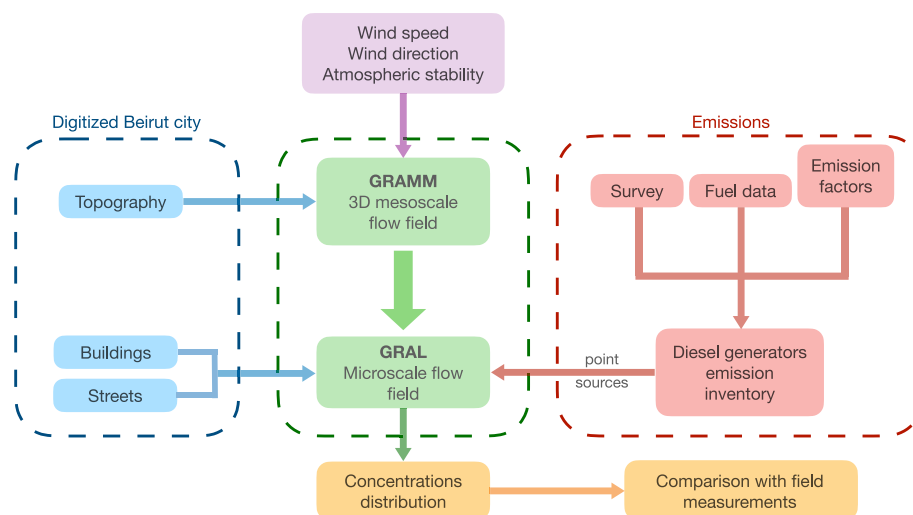


Fig. 1. Methodology scheme of the applied model and input data.

using several settings of meteorological data, emission patterns and urban features at two levels. The first level deals with general model properties including convergence, grid setup and computational behavior. The second one deals with test cases of modeled data against real-time measurements of wind data and pollutants concentrations. Both levels aim to prove the compliance of this model with the standard quality objectives and performance criteria. This was accomplished by different studies in the European countries and the model was found to be in compliance with the German Guideline VDI 3783 (2015) (Landesregierung, 2017, 2018) and to be fulfilling the criteria as defined by the Forum for Air quality Modelling in Europe (FAIR-MODE) (Berchet et al., 2017b). Another study, conducted in Australia, subjected GRAMM-GRAL to a model-to-data comparison as well as a model-to-model comparison. Both models were compared to commonly used models in Australia; GRAMM was compared to CALMET and GRAL was compared to CAL3QHCR, finding that GRAMM-GRAL produced results comparable to those of widely used local models (Roddiss et al., 2017a,b).

In this study, the GRAMM-GRAL tool is used to predict steady state-pollutant concentration distributions at a high spatial resolution (2 m and 5 m), under representative monthly meteorological conditions, taking into account the topography, building geometries, and their distribution, atmospheric stability, the spatial distribution of diesel generators, and their exhaust stack heights. For scenarios including building effects, GRAL's prognostic approach is used, where airflow is explicitly computed through forward integration of a set of prognostic equations. Fig. 1 illustrates the methodology used in this study to predict the steady-state spatial distribution of $PM_{2.5}$ concentrations representative of typical monthly meteorological conditions. The process began with digitizing Beirut City's topography, buildings, and street layout. Subsequently, an emission inventory for diesel generators was developed to serve as input for the dispersion modeling. The GRAMM module computes the mesoscale airflow field over a broader domain that includes GRAL's urban domain of interest. GRAMM takes as its input, in addition to the topography, wind speed (WSPD), wind direction (WD), and atmospheric stability class. Driven by GRAMM's outputs, GRAL computes the microscale airflow field within the city, resolving the effects of buildings on airflow and turbulence patterns. This microscale velocity field, combined with information on emission sources, is then used by GRAL's Lagrangian dispersion module to generate the pollutant concentration map in domain A. The spatial maps of $PM_{2.5}$ concentrations were generated using the graphical user interface (GUI) provided by the GRAMM-GRAL modeling system. This GUI facilitates visualization and processing of the model output, including the creation of detailed concentration distribution maps over the urban domain.

2.2. Domain setup for pollution transport simulations: Digitizing Beirut city

A digital representation of the topography, buildings, and streets of Beirut is provided in the Supplementary Material. The topography data, which was extracted at 50 m horizontal resolution for the Greater Beirut area (GPS, 2020), was used as an input to the GRAMM domain. Overlaid are 18478 buildings, provided as a shapefile by the Lebanese army, and the streets of Beirut downloaded as OSM files from OpenStreetMap, and then converted to a shapefile using Quantum Geographic Information System (QGIS). Note that the average building height and the standard deviation are 19 m and 12.88 m respectively.

In our implementation, the mesoscale airflow field is computed by GRAMM in domain B, while GRAL computations take place in domain A, which corresponds to Beirut city, as shown in Fig. 2-b. Domain A is used to study the dispersion of pollutants emitted by generators in Beirut at a 5 m resolution. As mentioned earlier, GRAMM requires a larger enclosing domain (domain B) to provide the velocity boundary conditions for domain A. The resolution of the GRAMM domain is set to 50 m and is bounded by the large yellow rectangle in Fig. 2-b. As seen in the topography color map in Fig. 2-a, Beirut is located near the center of a narrow coastal plain (210 km long and approximately 3 km wide) that stretches in a south-north direction. The city overlooks the Mediterranean Sea to the west and north. To the east, it is bounded by the western Mount Lebanon range, which extends nearly 170 km parallel to the Mediterranean coast, with peaks reaching approximately 3 km.

2.3. Diesel generators emission inventory

A detailed survey conducted by Baayoun et al. (2019) in the Hamra neighborhood of Beirut (yellow polygon labeled S in Fig. 2-b) collected data from 302 residential buildings, revealing that 53% had one or more diesel generators installed. The survey included information on generator location (indoor/outdoor), capacity, fuel consumption, and exhaust outlet height. This localized inventory was one of the few available bottom-up datasets for urban emissions in Beirut and helped highlight the widespread reliance on decentralized power generation in dense urban environments.

Based on that study, the distribution of the generators in the rest of the city considers the capacities of the generators in a statistical approach embracing the probability of a generator in a specific location to have a specific capacity (in kVA). In line with (Baayoun et al., 2019), we assumed a proportional relationship between generator capacity and fuel consumption. Their study categorized generators into small,

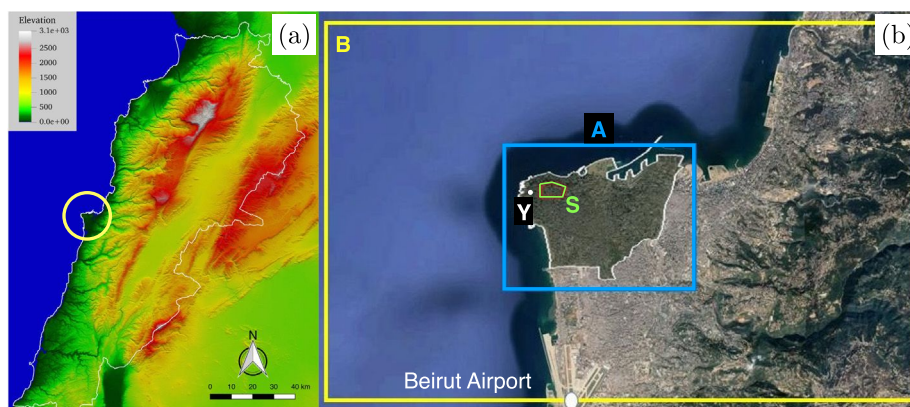


Fig. 2. (a) Color map topography of Lebanon. (b) GRAMM (yellow) and GRAL (blue) domains for Beirut city. Border for Beirut is delimited by the white line while the border for the Hamra neighborhood is delimited in green.

medium, and large, and used reported capacity and usage profiles to estimate fuel consumption rates per unit. Based on a detailed analysis of fuel distribution and generator surveys, they estimated that diesel generators in Beirut alone consumed approximately 747 tons of diesel fuel per day. This estimate was used in our modeling assumptions as a basis for calculating emissions. The $PM_{2.5}$ emission rate is then estimated using an average value of 1.2999 kg of $PM_{2.5}$ emitted per one metric ton of fuel consumption. This emission factor is based on Tier 1 default values provided by the EMEP/EEA 2016 Guidebook for small stationary diesel combustion sources, where values are expressed in units of grams per gigajoule (g/GJ) (EEA, 2016). To apply these factors to Beirut's diesel generator context, we converted them to kg per metric ton of fuel using a net calorific value consistent with the energy content of standard diesel. The resulting emission rate value aligns with recommended Tier 1 values for non-industrial oil-fired combustion in the absence of country-specific measurements (EEA, 2016).

To investigate the sensitivity of the $PM_{2.5}$ concentration spatial distribution to the generators locations, a second distribution of generators in Beirut was considered. Based on the survey in Hamra, this first order approximate distribution assumes one generator per two buildings in the rest of the city, making the number of diesel generators distributed in Beirut about 9369. As argued in the Supplementary Material, the assumption is justified, in part, due to the absence of availability of data on the various factors contributing to the generators spatial density. Since the exact location of these generators is not known outside Hamra, the generators were randomly placed at the centroids of the buildings footprints (GIS shapefile). The daily fuel consumption was distributed among individual generators randomly, while ensuring that the total citywide fuel use remained equal to 747 tons/day (Baayoun et al., 2019). For both distributions, all the diesel generators were assumed to operate for twelve hours per day. The two distributions, referred to as GD1 and GD2 respectively, are discussed in the Supplementary Material. As for the locations of the generator exhausts, two configurations were considered: (a) the exhaust stacks are positioned at the centroids of the buildings, 2 m above the rooftop (RT+2) and (b) the exhaust stacks are located at the street level, 4 m above the ground (G+4). Scenarios 1–8 listed in Table 1 investigate the effect of buildings, atmospheric stability and generators distribution, whereas scenarios 8–10 study the effect of generator exhaust stack heights, and scenarios 11–20 examine the effect of wind direction and speed.

2.4. Meteorological conditions

In addition to the emission sources, topography and building geometries, the dispersion model takes as input the wind speed, wind direction, and atmospheric stability conditions. Wind speed and direction data were obtained from Beirut Airport and averaged over

the period June 2005 to September 2018, using daily measurements recorded between 7:00 AM and 7:00 PM local time (WindFinder, 2019). The mean wind speed during this period was approximately 4 m/s (at the 10 m anemometer level). Based on this analysis, we identified the three most frequent wind directions for Beirut: 225° (southwest; from January to July and in December), 247° (west-southwest; August and September), and 337° (north-northwest; October and November). In our simulations, we considered wind speeds ranging from 1 m/s to 4 m/s to capture variability and account for low to moderate wind conditions typical of Beirut's urban environment. These wind directions and speeds were used to define representative meteorological scenarios for GRAMM-GRAL simulations. We emphasize that the model was not run over the full 2005–2018 period, but rather under these idealized, seasonally representative conditions. Atmospheric stability classes of “unstable” and “slightly unstable” were selected based on the Pasquill table (Berchet et al., 2017b). The different scenarios are summarized in Table 1.

2.5. Potential health implications

To assess the health impacts of air pollution more comprehensively, the WHO developed the AirQ+ tool (WHO, 2020). In this study, AirQ+ tool version 2.2 was used to estimate the health effects of air pollution based on concentration–response functions derived from epidemiological studies (WHO, 2018). The analysis is based on long-term exposure to $PM_{2.5}$ using total incidence from all natural causes for adults aged 30 and above, following WHO-recommended parameters in the AirQ+ methodology. Providing $PM_{2.5}$ concentration and population density inputs into AirQ+ allows estimating the potential health risks under different pollution scenarios. Hence, this tool facilitates the translation of pollutant concentration data into tangible health outcomes, providing a clearer picture of the public health implications of various air quality levels. Although no direct collaboration with an epidemiologist was undertaken, the analysis strictly followed the methodological guidelines provided by the WHO, including the appropriate selection of input data, concentration–response functions, and interpretation of results. Specifically, the default relative risk value for all-cause mortality associated with $PM_{2.5}$ exposure was set at 1.062 (95% CI: 1.040–1.083), and the WHO recommended cut-off concentration of 10 $\mu\text{g}/\text{m}^3$ was used to ensure consistency with international practices. AirQ+ has been utilized in numerous studies to assess the health impacts of air pollution across various regions and scenarios. For instance, a study employed AirQ+ to estimate the short- and long-term health effects and the attributable health risks due to multiple pollutants exposure in Kerman (Iran) (Malakootian and Mohammadi, 2020). Another study assessed the impact to human health and the burden of disease of various air pollutants, through the integration of AirQ+ with other statistical

Table 1

Settings of the investigated scenarios. U: unstable, SU: slightly unstable, GD1: statistical distribution of generators locations and capacities, GD2: first order approximate distribution of generators locations and capacities, G: ground level, RT roof top. ★ refers to the base case.

Scenario	Wind Speed (m/s)	Wind Direction (°)	Atmospheric Stability Class	Distribution of the generators	Exhaust elevation (m)	Buildings Included?
1	4	247	U	GD1	RT+2	N
2	4	247	U	GD1	RT+2	Y
3	4	247	SU	GD1	RT+2	N
4★	4	247	SU	GD1	RT+2	Y
5	4	247	U	GD2	RT+2	N
6	4	247	U	GD2	RT+2	Y
7	4	247	SU	GD2	RT+2	N
8★	4	247	SU	GD2	RT+2	Y
9	4	247	SU	GD1	G+4	Y
10	4	247	SU	GD2	G+4	Y
11	4	225	SU	GD1	RT+2	N
12	4	225	SU	GD1	RT+2	Y
13	4	337	SU	GD1	RT+2	N
14	4	337	SU	GD1	RT+2	Y
15	3	247	SU	GD1	RT+2	N
16	3	247	SU	GD1	RT+2	Y
17	2	247	SU	GD1	RT+2	N
18	2	247	SU	GD1	RT+2	Y
19	1	247	SU	GD1	RT+2	N
20	1	247	SU	GD1	RT+2	Y
21	1	247	SU	GD1	G+4	Y
22	4	247	SU	GD2	G+4	Y

models over the Camp de Tarragona County (Catalonia, Spain) (Rovira et al., 2020).

The data for population density, which is an input to the AirQ+ software, is obtained from Data for Good by Meta (DataForGood, 2024). This dataset was generated using a machine-learning based computer vision method that creates population maps from satellite imagery at a global scale, with spatial sensitivity corresponding to individual buildings and suitable for worldwide deployment. By combining this settlement data with census data, the population maps are created with 30 m resolution for 18 countries (Tiecke et al., 2017). The population density distribution in the city of Beirut is provided in the Supplementary Material. Population density in Beirut is highest in the southwestern and northeastern areas, while lower population densities are observed in the northern and southeastern parts of the city.

PM_{2.5} concentrations at 5 m × 5 m resolution were obtained from GRAMM-GRAL simulations, while population density data at 30 m × 30 m resolution were sourced from Data for Good by Meta. To spatially align the two datasets, the k-d tree method was applied to assign each PM_{2.5} grid cell its nearest population density value. Then, using the WHO recommended log-linear concentration–response function, the number of attributable mortality cases was estimated at cell level. This high resolution approach enable to account for spatial heterogeneity in both air pollution exposure and population distribution, leading to a more detailed and spatially accurate estimation of health impacts.

2.6. Satellite-based PM_{2.5} patterns for visual comparison

To provide qualitative regional context, we included a visual comparison between the modeled dispersion results and annual average satellite-derived PM_{2.5} data for the same study region. The satellite product has a high-resolution of (0.01° × 0.01°) and is based on PM_{2.5} monthly datasets from the Atmospheric Composition Analysis Group of the Washington University in St. Louis (Shen et al., 2024). This group estimated the global PM_{2.5} concentrations using data from satellites, simulations, and monitor-based sources. Aerosol optical depth readings were gathered from multiple satellites (NASA MODIS, MISR, SeaWiFS, and VIIRS), and their respective retrievals (DARK Target, Deep Blue,

MAIAC) were combined with simulations from the GEOS-Chem chemical transport model to establish the geophysical relationship between aerosol optical depth and surface PM_{2.5}.

Initial PM_{2.5} estimates from satellite and chemical transport models were calibrated using a convolutional neural network trained on ground-based observations from major global networks, including the U.S. EPA, European Environment Agency (EEA), World Health Organization, and OpenAQ. Rather than directly interpolating sparse ground data, the model learned the spatial bias between geophysical PM_{2.5} estimates and observed concentrations, using high-resolution inputs such as aerosol optical depth, meteorology, and land use. These features, available globally at 0.01° resolution, allowed the model to apply learned correction patterns even in unmonitored regions. In these regions, a geographically weighted averaging scheme was used to blend bias-corrected predictions with the raw geophysical estimates. To further improve stability in data-sparse regions, the model's loss function was designed to nudge predictions toward geophysical estimates, avoiding unrealistic results where observations are lacking. It is important to note that this product involves multiple layers of modeling, including smoothing by convolutional neural networks and GEOS-Chem simulations, and should not be interpreted as a ground-truth observation. This comparison is used purely for illustrative, pattern-level insights, not for validation.

3. Results and discussion

3.1. Comparison with satellite images

Fig. 3 presents a qualitative comparison between satellite-derived PM_{2.5} concentration maps (left column) and model-predicted PM_{2.5} distributions (right column) for three prevailing wind directions. This comparison is intended solely as a visual pattern check to assess the broad spatial distribution similarities, rather than a quantitative validation or benchmark of model performance. Certain large-scale spatial trends of PM_{2.5} coverage can be observed when comparing the model-predicted images to the satellite images. A northward shift in the PM_{2.5} dispersion pattern was noticed for the southwest (225°) and

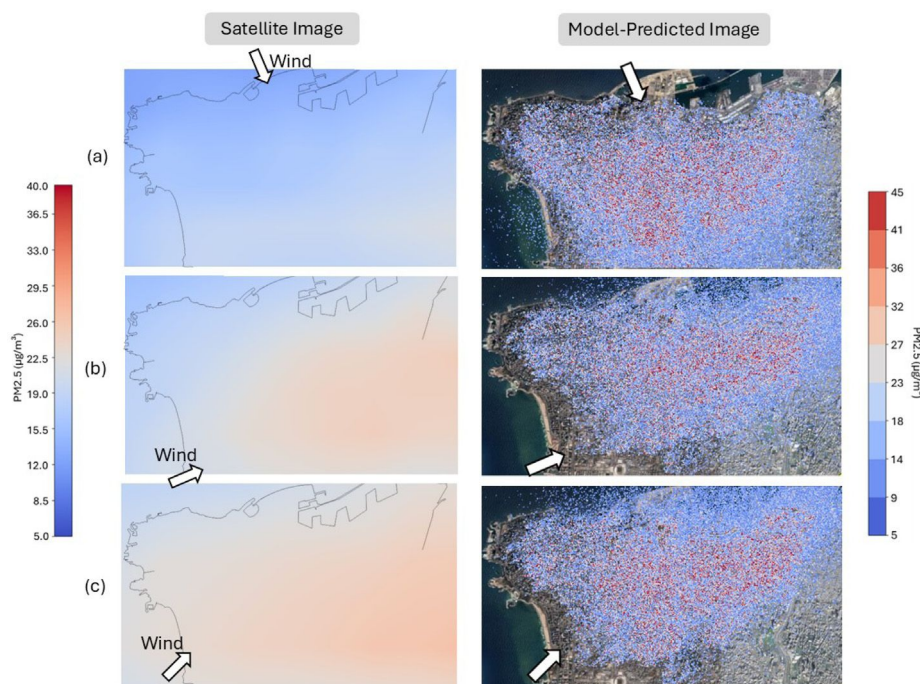


Fig. 3. Comparison of Satellite Images and Model-Predicted Images of $PM_{2.5}$ Distribution in Beirut at three different wind directions with respect to a reference point: (a) north-northwest (337°), (b) west-southwest (247°), and (c) southwest (225°).

the west-southwest (247°) wind directions in both satellite and model-predicted images. In the case of north-northwest wind direction (337°), a southward shift in the $PM_{2.5}$ dispersion pattern was observed in the model-predicted images which aligns with the satellite images. $PM_{2.5}$ hotspots were also noticed downstream of the wind direction in both the model-predicted and satellite images.

It is worth noting the inherent differences in data sources and processing methods, which introduce complexities that limit the validity of quantitative comparison. Notably, the vertical correction applied to satellite data used to estimate ground-level $PM_{2.5}$ concentration can introduce discrepancies. This correction is influenced by several factors, including atmospheric mixing, boundary layer height, and aerosols vertical profiles, which may not be fully captured or aligned with model assumptions. Moreover, the presence of additional $PM_{2.5}$ sources, beyond the modeled diesel generator emissions, such as industrial activities, vehicular emissions, and natural sources like dust, further complicated direct quantitative validation. In addition, the significant difference in spatial resolution between the two datasets, around 1 km for satellite-derived $PM_{2.5}$ versus 5 m for GRAMM-GRAL, can lead to substantial differences in predicted concentrations. The coarse satellite resolution inherently smooths fine-scale spatial variability, potentially underrepresenting peak concentrations near localized sources. Given these factors, this visual pattern comparison does not constitute model validation or proof of dispersion accuracy. Rather, it serves as an initial qualitative assessment of consistency in large-scale $PM_{2.5}$ distribution patterns between the modeled and satellite-derived data, and aims to contextualize GRAMM-GRAL predictions against existing regional estimates.

3.2. Effect of topography

The topography plays a key role in the large-scale pollution transport. This is because the pollutant is advected by the velocity field, which in turn depends on the topography. Based on the law of mass conservation, one expects high wind speeds on elevated terrains such as hills and low wind speeds in the valleys (Oke, 2002). In the absence

of obstacles, the pollutant residence time is largely decided by the wind speed so that the pollutant gets swept away quickly from regions of high wind speed and tends to accumulate in regions where the wind speed is low (Whiteman, 2000).

Beirut's topography, bordered by the Mediterranean Sea to the west and Mount Lebanon to the east, includes two hills about 100 m high separated by a lower-elevation pass. The resulting velocity field shows that lower speeds are observed in the east side of the city, which is characterized by a downward slope created by the river of Beirut that separates the district of Matn in the Mount Lebanon Governorate from Beirut. Higher wind speeds are observed in the western side, especially in at the north-northwest near the sea. By considering only the incoming wind conditions and the topography, pollutant concentrations tend to be higher in the low wind speed eastern areas and lower in the windward western regions where ventilation is enhanced (Beychok, 1994). A visual representation of the wind streamlines and velocity field is provided in the Supplementary Materials.

3.3. Effect of buildings

The spatial distributions of the $PM_{2.5}$ concentration at 2 m above the ground for the base case (scenario 4) is shown in Fig. 4-b. For this scenario, the buildings are included, the wind speed and direction are 4 m/s and west-southwest (247°) respectively, the atmospheric stability class is "slightly unstable", and the generators distribution is GD1 with exhaust stacks located at 2 m above rooftops. Fig. 4-a shows the $PM_{2.5}$ concentration map for scenario 3, where all conditions are the same as the base case except that buildings are not included. When comparing the two distributions, one can observe that the presence of the buildings results in a highly non-uniform distribution characterized by many islands of elevated $PM_{2.5}$ concentration. The presence of the buildings is also responsible for 27% rise in estimated health risk, as inferred from the lower value of potential health impacts in scenario 13. The presence of the buildings resulted in around 25% increase in $\overline{PM}_{2.5}^{(25\text{ m})}$ (2 m) the average $PM_{2.5}$ concentration, which is close to 25%, the reduction in non-built up area brought about by adding the buildings.

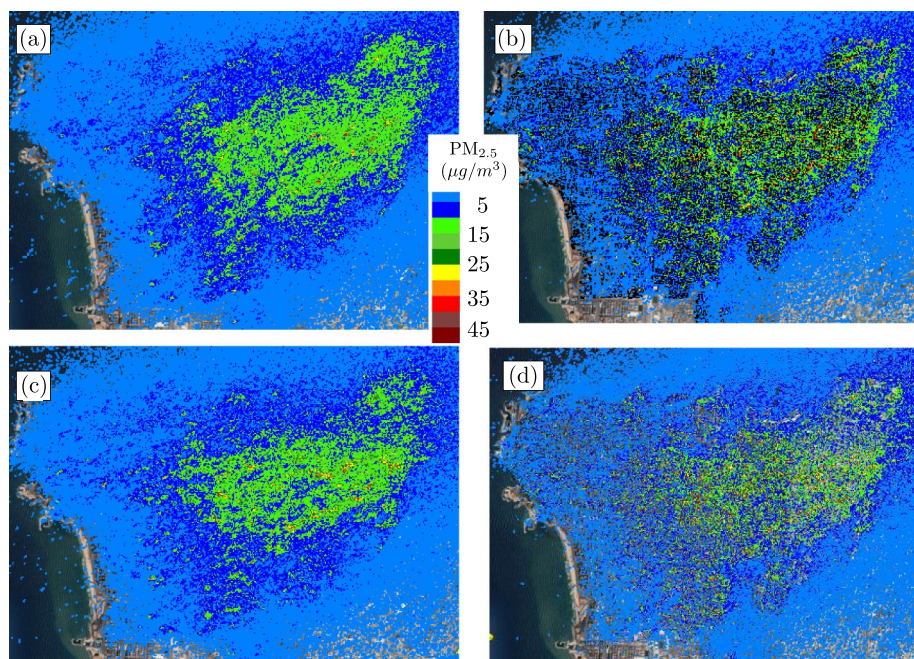


Fig. 4. $PM_{2.5}$ concentration maps at 2 m above ground level for (a) no buildings, slightly unstable atmosphere (scenario 3), (b) with buildings, slightly unstable atmosphere (scenario 4), (c) no buildings, unstable atmosphere (scenario 1) and (d) with buildings, unstable atmosphere (scenario 2).

3.4. Effect of atmospheric stability class

The impact of the atmospheric stability class can be discerned by comparing scenarios 2 and 4 (with buildings) and scenarios 1 and 3 (no buildings). In the presence of buildings, the spatial distributions of $PM_{2.5}$ concentrations at 2 m above the ground for the slightly unstable (scenario 4) and unstable (scenario 2) cases are shown in Figs. 4-b and d under steady-state conditions. While both distributions reflect similar overall plume extents, scenario 4 exhibits more distinct clusters of elevated $PM_{2.5}$ concentrations in the 40–45 $\mu\text{g}/\text{m}^3$ range, particularly near source-dense areas. This may be attributed to reduced turbulent dispersion under slightly unstable conditions, allowing for more localized accumulation. In the absence of buildings, the $PM_{2.5}$ concentration maps are shown in Figs. 4-a and c for the slightly unstable (scenario 3) and unstable (scenario 1) cases. Overall, scenarios 1 and 3 show similar trends. However, under slightly unstable conditions (scenario 3), the plume appears more stretched along the wind direction and exhibits sharper spatial gradients, reflecting differences in atmospheric mixing.

3.5. Effect of wind speed and direction

The wind conditions are among the main meteorological conditions that affect the dispersion of $PM_{2.5}$ and other pollutants. The impact of the wind direction on the $PM_{2.5}$ map is shown in Fig. 5 for the following prevailing wind directions: southwest (225°; January to July and December), west-southwest (247°; August and September), and north-northwest (337°; October and November). The figures on the left are for scenarios without buildings and those on the right are for scenarios with buildings. As expected, we notice spatial variation of the pollutant dispersion along the direction of the downstream wind. Since the wind direction differs by only 22°, the large scale patterns in Figs. 5-a and c (scenarios 11 and 3) and Figs. 5-b and d (scenarios 12 and 4) are similar. For the scenarios (13 and 14) where the wind direction is north-northwest (337°), we notice a southward shift in the dispersion pattern with clear separation between an eastern and a western zone along the shallow valley between the two hills (see Figure S.14 in the Supplementary Material). The impact of the wind

direction on the spatial distribution of potential health risks can be seen in Figs. 6-a, b, and c for $WD=337^\circ$ (north-northwest), $WD=247^\circ$ (west-southwest), and $WD=225^\circ$ (southwest). The distributions are over the latitude-longitude plane of Beirut at 2 m elevation. By closely examining these maps, we notice that a change in wind direction significantly alters the locations of zones with elevated potential health risks. However, these risk hotspots consistently shift downstream, following the pattern of pollutant dispersion. To enhance the visibility of localized health risks, a zoomed-in view of zone Z1 is provided in Figure S.15 in the Supplementary Material, illustrating the spatial distribution of estimated potential health risks in greater detail. The percentage of estimated health impacts is highly heterogeneous within Z1. The highest-risk areas correspond to locations with both high modeled $PM_{2.5}$ concentrations and dense population, reinforcing the need for localized intervention strategies in similar urban microenvironments.

The values of the normalized mean concentrations of $PM_{2.5}$ at 2 m elevation and the estimated health impact metrics for the considered wind direction and wind speed are presented in Figs. 7-a and d. The normalized $PM_{2.5}$ mean concentrations at 2 m elevation, plotted in Figs. 7-a and b, show that they are roughly the same for all the scenarios without buildings (3, 11, 13) with $\overline{PM}_{2.5}(2\text{ m}) = 5.79 - 6.03 \mu\text{g}/\text{m}^3$ and for the scenarios with buildings (4, 12, 14) with $\overline{PM}_{2.5}(2\text{ m}) = 7.22 - 7.45 \mu\text{g}/\text{m}^3$. The estimated potential health impacts, plotted in Figs. 7-c and d, are the largest when the wind direction is north-northwest (337°) which may be explained by the fact that highly populated areas are observed downwind (see Figure S.12 in the Supplementary Material).

The impact of the wind speed on the $PM_{2.5}$ concentration map is shown in Fig. 8 for the following wind speeds: 1 m/s (Figs. 8-a and b), 2 m/s (Figs. 8-c and d), 3 m/s (Figs. 8-e and f), and 4 m/s (Figs. 8-g and h). These predictions are for a slightly unstable atmosphere and the prevailing west-southwest wind direction (247°). The figures on the left are for the scenarios without buildings and those on the right are for the scenarios with buildings. We notice that as the wind speed decreases, both the intensity and the spatial coverage of the $PM_{2.5}$ concentration distribution increases. The estimated mortality burden strongly depends on the wind speed, with higher values observed at

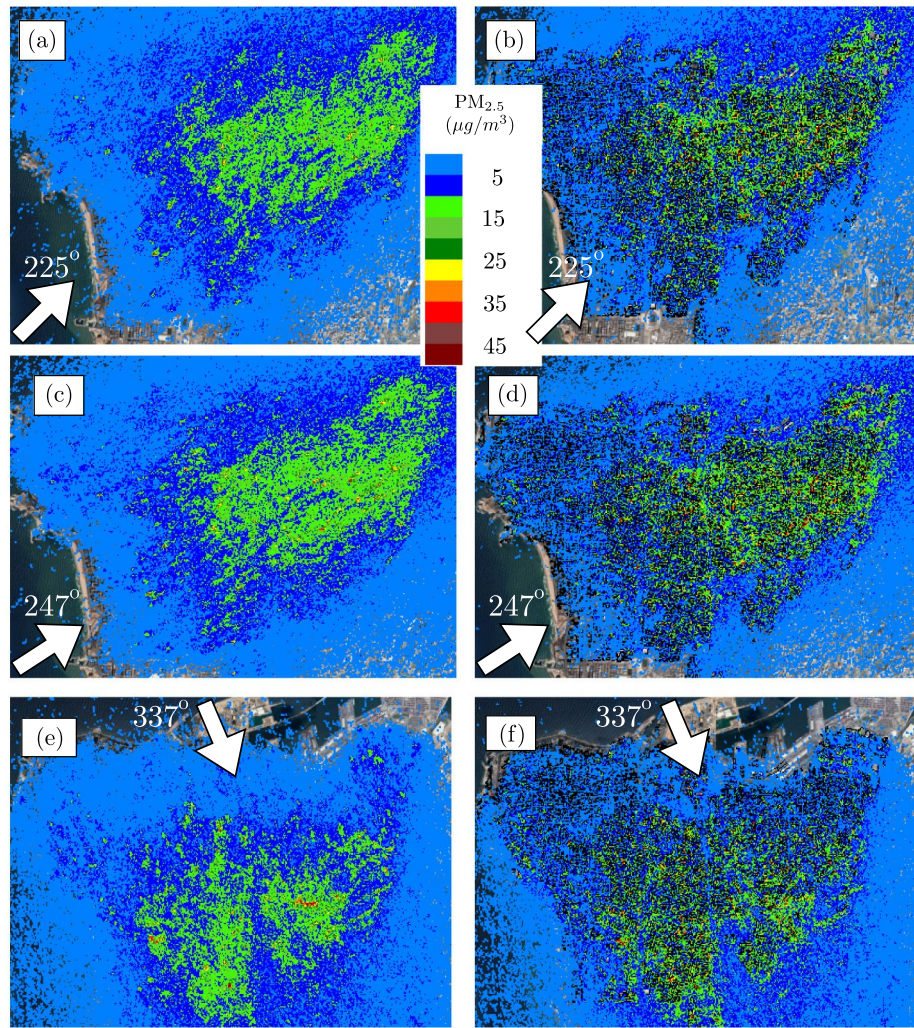


Fig. 5. $PM_{2.5}$ concentration maps at 2 m above ground for (a) southwest wind direction (225°) without buildings (scenario 11) (b) southwest wind direction (225°) with buildings (scenario 12) (c) west-southwest wind direction (247°) without buildings (scenario 3) (d) west-southwest wind direction (247°) with buildings (scenario 4) (e) north-northwest wind direction (337°) without buildings (scenario 13) (f) north-northwest wind direction (337°) with buildings (scenario 14). Solutions are for a slightly unstable atmosphere at 4 m/s wind speed.

lower wind speeds. The impact of the wind speed on $\overline{PM}_{2.5}$ (2 m), and on the estimated potential health risks is shown in Figs. 7- b and d respectively. As the wind speed decreases from 4 m/s to 1 m/s, the estimated mortality burden and $PM_{2.5}$ (2 m) increase by 383% and 208%, respectively, in the presence of buildings. For low wind speeds (≤ 3 m/s), localized highly polluted zones (similar to zone Z1 observed in scenario 16 of Fig. 8-f) start to appear, especially for the scenarios where the buildings are present (Figs. 8-b, d, f, and h). The number and intensity of these zones increase as the wind speed decreases. These observations are a reflection of the impact of the wind speed, which cannot be overstated. Higher wind speeds enhance the ventilation of pollutants while lower wind speeds promote accumulation of pollutants. Low wind speeds, together with short street canyons that are oriented perpendicular to the incoming wind direction are the ingredients that make up highly polluted zones. Notice that for the case of diesel generators, short street canyons imply that the emission sources (within the canyon) are necessarily located near the ground.

3.6. Effect of generator exhaust stack heights

Figs. 9-a and c show the spatial distributions $PM_{2.5}$ concentration at 2 m above the ground for scenarios 4 and 9. While these two scenarios

share the same wind speed (4 m/s) and southwest direction (225°), atmospheric stability class (slightly unstable), generators distribution in the longitude-latitude plane (GD1), they differ in the exhaust stack heights. Upon placing the emission sources at the street level (4 m above ground), the map shown in Fig. 9-c shows that a larger area is affected with the presence of many more hot spots as compared to the case when the generator exhaust stacks are at 2 m above the rooftops (Fig. 9-a). Accordingly, the mean $PM_{2.5}$ concentration at 2 m increase by 70% when the emissions source are brought down to the street level. Similar observations can be made for corresponding two scenarios using generators distribution GD2, which randomly allocates one generator per two buildings (refer to Section 2.3 for more details). The $PM_{2.5}$ concentration maps of scenarios 8 and 10 are shown in Figs. 9-b and d.

Fig. 7-e shows a plot of the potential health versus wind speed for the generator exhaust stack heights: 2 m above rooftop (RT+2) and 4 m above ground level (G+4). The plot shows that placing the exhaust stacks on street level instead of roof tops does not have as much impact at lower wind speeds. Placing the exhaust stack at 4 m above the ground results in 622 additional cases of adverse health impacts, equivalent to a 2% increase when the wind speed is 1 m/s, as opposed to 2455 additional cases, representing a 28% increase when

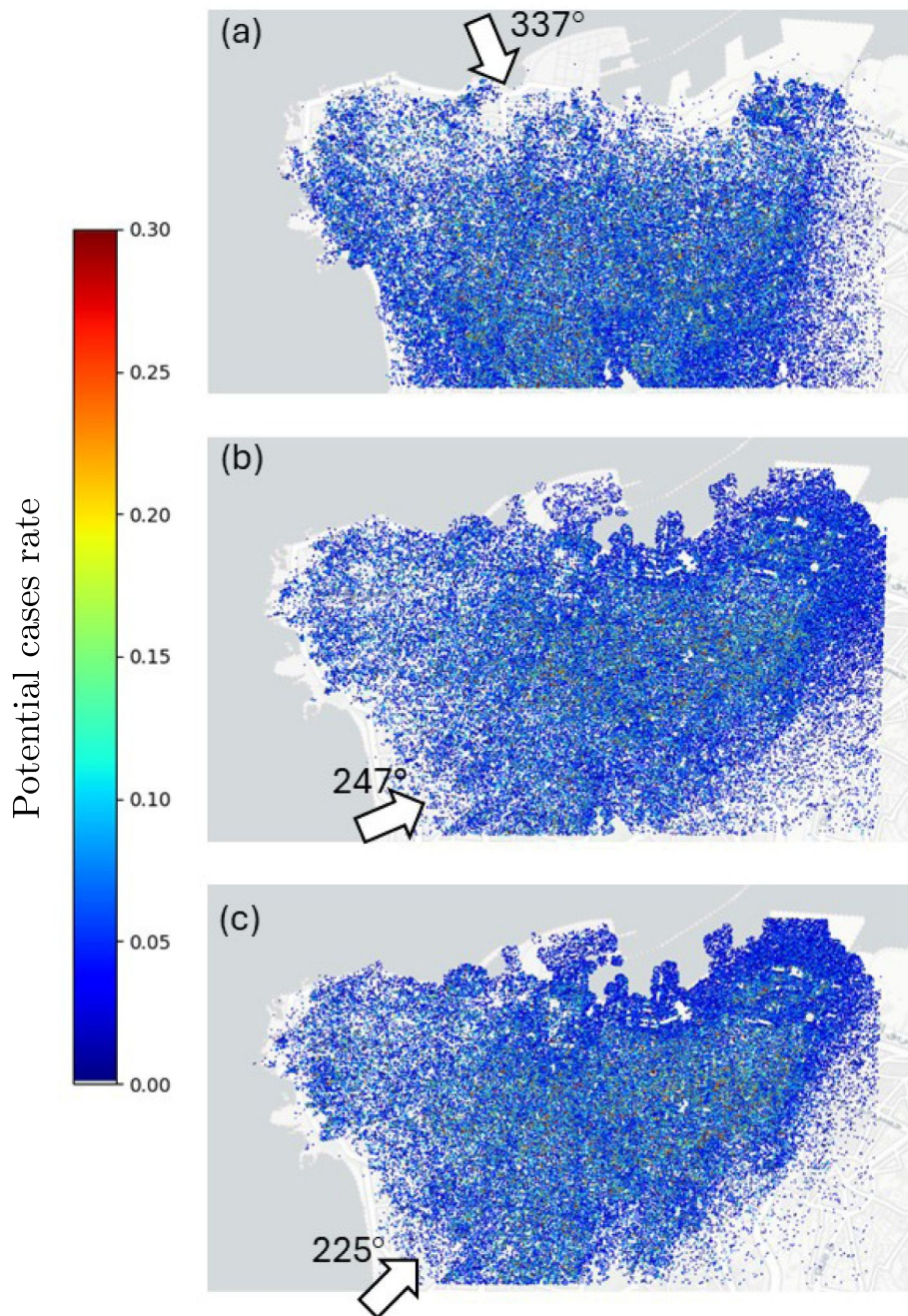


Fig. 6. Spatial distribution of estimated potential health risks related to $PM_{2.5}$ exposure over Beirut for three prevailing wind directions (a) north-northwest (337°), (b) west-southwest (247°), and (c) southwest (225°).

the wind speed is 4 m/s. We deduce that when the prevailing wind speed decreases, the exposure becomes less sensitive to the exhaust stack heights. Given these results, increasing the generator exhaust stack height is an effective mitigation measure only at high enough wind speeds.

Upon inspecting Figs. 9-a and b, one can observe that the two generators distributions (GD1 and GD2) yield similar large-scale patterns of $PM_{2.5}$ dispersion, implying that the dispersion pattern is, to a large degree, insensitive to the exact location of the generator exhaust stacks, subject to the constraints governing the underlying statistical distributions of their longitude–latitude locations. Upon placing the emission sources at the street level (4 m above ground), the maps shown in Figs. 9-c and d, exhibit similar patterns on the large scale.

One can, however, observe the existence of localized highly polluted zones (e.g. zone Z1) in scenarios 8 and 10 (Figs. 9-b and d). Comparison with Figs. 9-a and c shows that the pollution dispersion in such zones is highly sensitive to the locations of the emission sources and their emission rates. These zones are commonly found in regions of low velocity, high buildings density, and street canyons perpendicular to the wind direction. An illustration of such regions is provided in the Supplementary Material. In Zone Z1, the canyon is oriented perpendicular to the wind direction, pockets of pollution are observed. According to the ratios of building height to street width and building length to height, these canyons are classified as deep narrow canyons where the most circular eddy emerges in the skimming flow regime (Oke, 1988). These observations are in line with what is reported in the literature

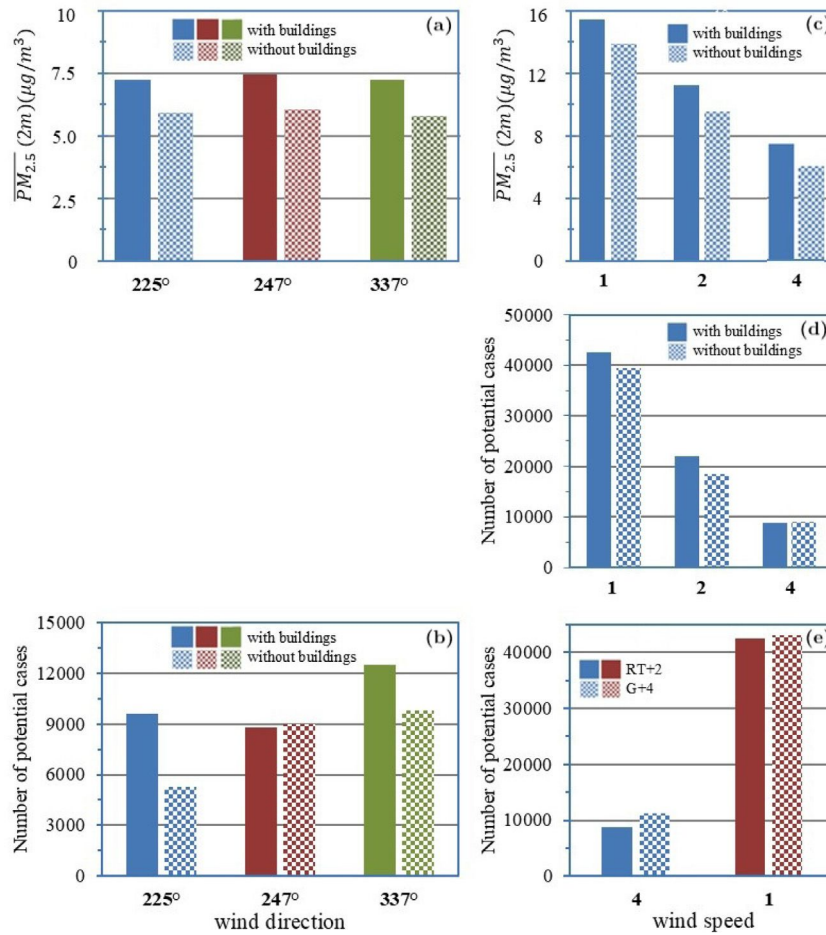


Fig. 7. (a) mean ground level $PM_{2.5}$ concentration versus wind direction, (b) estimated number of deaths attributed to $PM_{2.5}$ exposure versus wind direction, (c) mean ground level $PM_{2.5}$ concentration versus wind speed, (d) estimated number of deaths attributed to $PM_{2.5}$ exposure versus wind speed and buildings configurations, (e) estimated number of deaths attributed to $PM_{2.5}$ exposure versus wind speed and generator exhaust stack heights. Unless otherwise specified, the following conditions apply: slightly unstable atmosphere, wind direction: west-southwest (247°), wind speed: 4 m/s, generator exhaust stack heights: 2 m above rooftops.

under similar conditions (Afiq et al., 2012; Yazid et al., 2014). On the other hand, when the street canyon is parallel to the predominant wind direction, the wind is accelerated through the canyon driving the pollution away from the canyon.

4. Conclusion and future work

In this work, we carried out a model-based study of $PM_{2.5}$ dispersion originating from a large number of diesel generators installed in Beirut. In particular, the spatial distribution of the $PM_{2.5}$ concentration, predicted at steady state using the computational modeling tool GRAMM-GRAL for Lagrangian transport, was utilized to explore large scale dispersion patterns over the entire city. Several scenarios were considered to investigate the impact of topography, atmospheric stability, presence of buildings, diesel generators distribution, and emission sources elevations (exhaust stacks heights) for representative meteorological conditions of the wind speeds and directions.

The simulations showed that the large scale dispersion patterns are strongly influenced by the wind speed and direction. While the wind direction shift the entire distribution along the wind direction, smaller wind speeds increases the intensity and expands the spatial coverage of the $PM_{2.5}$ concentration. For the case where the generator exhaust stacks are located at 2 m above rooftops and for west-southwest wind direction (247°), the simulations showed that reducing the wind speed

from 4 m/s to 1 m/s resulted in more than two fold increase in the mean concentration at street level and the potential health risks. For a given wind direction, and atmospheric stability condition, moving the generator exhaust stacks from street level to the top of the buildings resulted in significant reduction in the ground level concentration distribution at high speeds. At lower speeds, this reduction was less which limits the effectiveness of elevating the generator exhaust stacks to rooftops in mitigating street level exposure.

These findings have important public health implications. Populations living in low-wind areas with low stack heights are at increased risk of prolonged $PM_{2.5}$ exposure. Our results suggest that local authorities could consider modifying building codes to require higher emission release heights or implementing urban design strategies to enhance natural ventilation. Moreover, this modeling framework can support epidemiological studies by identifying exposure hotspots, enabling more accurate health burden estimations, and informing risk mapping efforts. It may also assist policymakers in planning targeted interventions and designing cleaner, healthier urban environments through scenario testing and spatial optimization of emission controls.

By accounting for wind conditions, topography, urban geometries, emission distributions, and population density, this physically grounded approach offers a valuable tool for assessing exposure and shaping evidence-based environmental and public health policies.

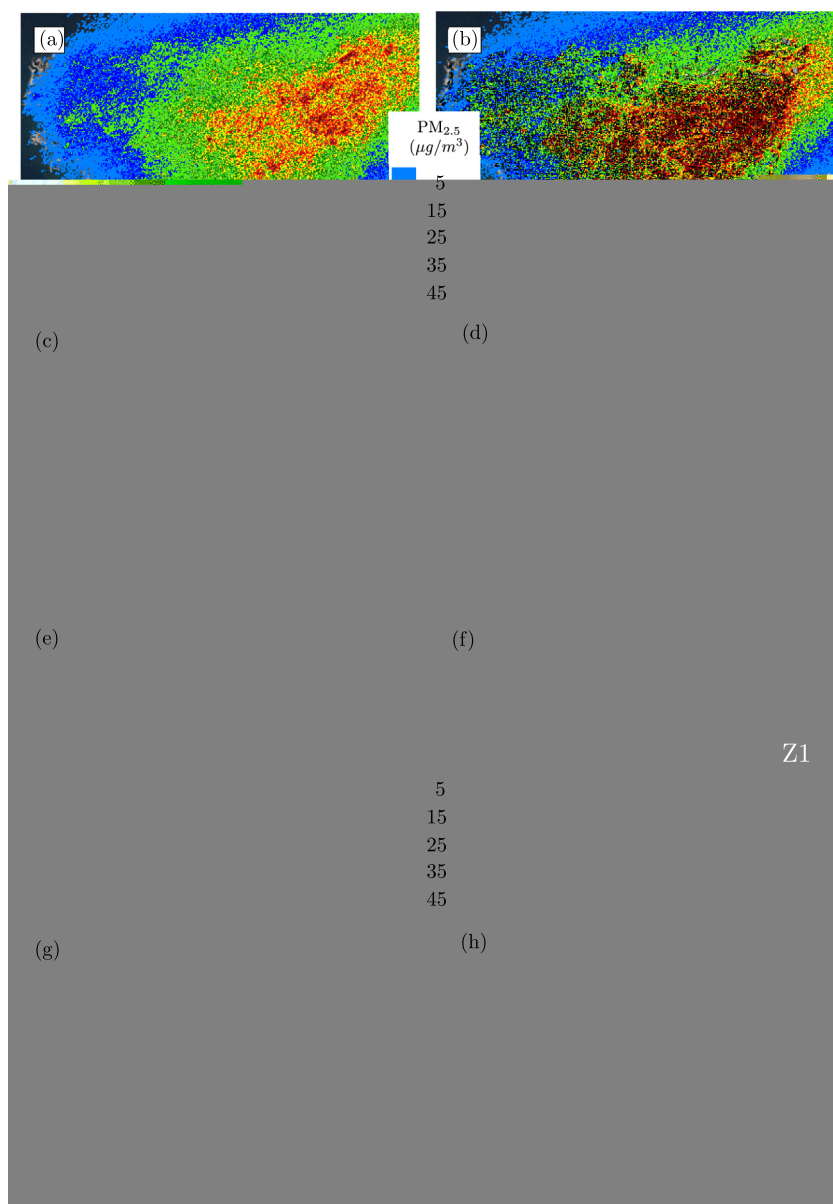


Fig. 8. PM_{2.5} concentration maps at 2 m above ground for (a) wind speed 1 m/s without buildings (scenario 19) (b) wind speed 1 m/s with buildings (scenario 20) (c) wind speed 2 m/s without buildings (scenario 17) (d) wind speed 2 m/s with buildings (scenario 18) (e) wind speed 3 m/s without buildings (scenario 15) (f) wind speed 3 m/s with buildings (scenario 16) (g) wind speed 4 m/s without buildings (scenario 3) (h) wind speed 4 m/s with buildings (scenario 4). Solutions are for a slightly unstable atmosphere and west-southwest wind direction (247°).

CRediT authorship contribution statement

Elissar Al Aawar: Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Jennifer Karam:** Writing – review & editing, Visualization, Software, Methodology, Investigation. **Abdelkader Baayoun:** Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Mohamad Ghadban:** Writing – review & editing, Software, Methodology, Investigation, Conceptualization. **Sara Najem:** Writing – review & editing, Validation, Resources, Investigation, Data curation. **Najat A. Saliba:** Writing – review & editing, Validation, Supervision, Investigation, Formal analysis, Conceptualization. **Issam Lakkis:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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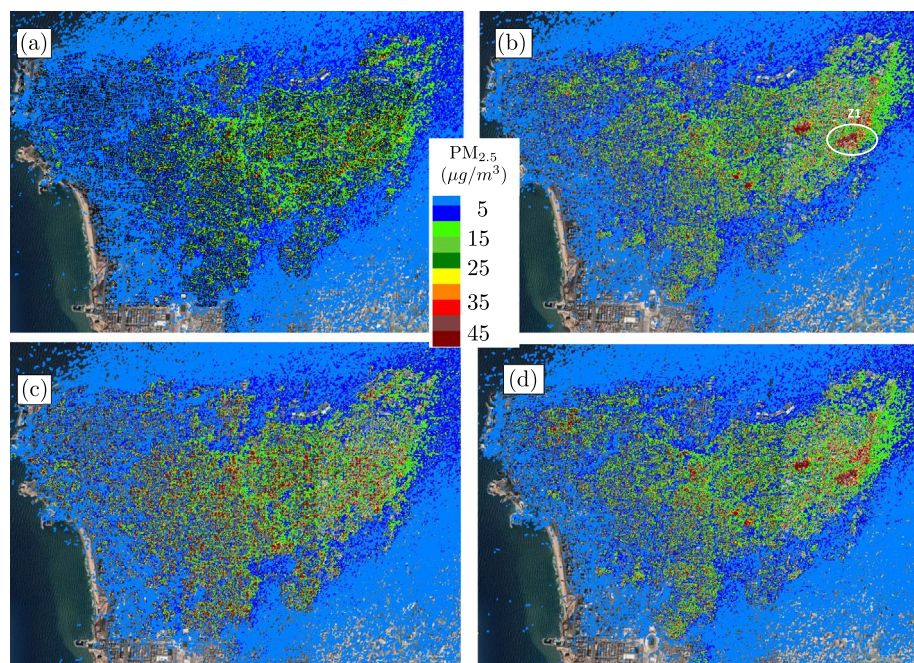


Fig. 9. $PM_{2.5}$ concentration maps at 2 m above ground for (a) scenario 4, (b) scenario 8, (c) scenario 9, and (d) scenario 10. Solutions are for a slightly unstable atmosphere.

Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.apr.2025.102686>.

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